
Redyx

Feature Completion Tracker

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1 Feature Completion Tracker

This document lists all features and requirements promised in the Redyx Software Requirement Specification (SRS) and tracks their completion status.

1.1 1. Functional Requirements

1.1.1 1.1 Authentication (FR-AUTH)

ID	Requirement	Priority	Status
FR-AUTH-1	Users can register with email, username, and password	P0	Completed
FR-AUTH-2	Users can register via Google OAuth, then choose a username	P0	Completed
FR-AUTH-3	Email is verified through a 6-digit OTP before account activation	P0	Completed
FR-AUTH-4	Users can log in with email/password or Google OAuth	P0	Completed
FR-AUTH-5	The system issues a short-lived access token (15 min) and a refresh token (7 days)	P0	Completed
FR-AUTH-6	Users can log out, which invalidates the refresh token	P0	Completed
FR-AUTH-7	Passwords are hashed with <code>argon2id</code> before storage	P0	Completed

ID	Requirement	Priority	Status
FR-AUTH-8	Email and auth method are never exposed to other users	P0	Completed

1.1.2 1.2 User Profiles (FR-USER)

ID	Requirement	Priority	Status
FR-USER-1	Each user has a public profile showing username, karma, and cake day	P1	Completed
FR-USER-2	The profile displays the user's post and comment history	P1	Completed
FR-USER-3	Karma is calculated from the total upvotes received on posts and comments	P1	Completed
FR-USER-4	Users can update their display settings and avatar	P2	Completed
FR-USER-5	Users can delete their account, which wipes all PII and replaces their posts/comments with [deleted]	P1	Completed

1.1.3 1.3 Communities (FR-COMM)

ID	Requirement	Priority	Status
FR-COMM-1	Any authenticated user can create a community with a unique name	P0	Completed
FR-COMM-2	Community names are immutable once created	P0	Completed

ID	Requirement	Priority	Status
FR-COMM-3	Communities have a description, rules, banner, and icon	P1	Completed
FR-COMM-4	Communities can be set to public, restricted, or private visibility	P1	Completed
FR-COMM-5	Users can join and leave communities	P0	Completed
FR-COMM-6	The creator of a community is automatically a moderator	P0	Completed
FR-COMM-7	Moderators can assign other users as moderators	P1	Completed

1.1.4 1.4 Posts (FR-POST)

ID	Requirement	Priority	Status
FR-POST-1	Users can create text posts (title + markdown body) in a community	P0	Completed
FR-POST-2	Users can create link posts (title + URL)	P1	Completed
FR-POST-3	Users can create image/video posts (title + media upload)	P2	Completed
FR-POST-4	Each post belongs to exactly one community	P0	Completed
FR-POST-5	Posts display: title, author, community, timestamp, vote score, comment count	P0	Completed
FR-POST-6	Users can edit and delete their own posts	P0	Completed
FR-POST-7	The home feed aggregates posts from all communities the user has joined	P1	Completed
FR-POST-8	Posts can be sorted by Hot, New, Top (with time filter), and Rising	P1	Completed

ID	Requirement	Priority	Status
FR-POST-9	Users can optionally post as [anonymous] within a community	P2	Completed

1.1.5 1.5 Comments (FR-CMNT)

ID	Requirement	Priority	Status
FR-CMNT-1	Users can comment on posts	P0	Completed
FR-CMNT-2	Users can reply to comments, forming nested threads	P0	Completed
FR-CMNT-3	Comments display: author, timestamp, vote score, reply count	P0	Completed
FR-CMNT-4	Comments can be sorted by Best, Top, New, Controversial	P1	Completed
FR-CMNT-5	Deleted comments show [deleted] but the thread structure remains intact	P0	Completed
FR-CMNT-6	Deep threads are lazy-loaded (top 2-3 levels shown, rest on demand)	P1	Completed

1.1.6 1.6 Voting (FR-VOTE)

ID	Requirement	Priority	Status
FR-VOTE-1	Users can upvote or downvote any post or comment	P0	Completed
FR-VOTE-2	Each user gets one vote per item; they can change or remove it	P0	Completed
FR-VOTE-3	Net score (upvotes minus downvotes) is displayed on each item	P0	Completed
FR-VOTE-4	Votes update the author's karma asynchronously via Kafka	P1	Completed

ID	Requirement	Priority	Status
FR-VOTE-5	Vote endpoints are idempotent (duplicate requests are safe)	P0	Completed

1.1.7 1.7 Search (FR-SRCH)

ID	Requirement	Priority	Status
FR-SRCH-1	Users can search posts by title and body text	P2	Completed
FR-SRCH-2	Users can search within a specific community or globally	P2	Completed
FR-SRCH-3	Community name autocomplete is available in the search bar	P2	Completed
FR-SRCH-4	Search results are ranked by relevance, recency, and vote score	P2	Completed

1.1.8 1.8 Notifications (FR-NOTF)

ID	Requirement	Priority	Status
FR-NOTF-1	Users receive a notification when someone replies to their post or comment	P1	Completed
FR-NOTF-2	Users receive a notification when mentioned with u/username	P2	Completed
FR-NOTF-3	Notifications are delivered in real time via WebSocket	P1	Completed
FR-NOTF-4	If the user is offline, notifications are stored and delivered on reconnect	P1	Completed
FR-NOTF-5	Users can mark notifications as read	P1	Completed
FR-NOTF-6	Users can configure notification preferences (mute communities, mute replies)	P2	Completed

1.1.9 1.9 Media (FR-MDIA)

ID	Requirement	Priority	Status
FR-MDIA-1	Users can upload images and videos when creating a post	P2	Completed
FR-MDIA-2	Uploaded files are validated for type and size before storage	P2	Completed
FR-MDIA-3	Thumbnails are generated for image uploads	P2	Completed
FR-MDIA-4	Media is stored in AWS S3 and served through CloudFront CDN	P2	Completed

1.1.10 1.10 Moderation (FR-MOD)

ID	Requirement	Priority	Status
FR-MOD-1	Moderators can remove posts and comments from their community	P2	Completed
FR-MOD-2	Moderators can ban users from their community (with duration and reason)	P2	Completed
FR-MOD-3	Moderators can pin up to 2 posts in their community	P2	Completed
FR-MOD-4	All moderation actions are recorded in a mod log	P2	Completed
FR-MOD-5	Moderators can view a queue of reported/flagged content	P2	Completed

1.1.11 1.11 Rate Limiting (FR-RATE)

ID	Requirement	Priority	Status
FR-RATE-1	The API gateway enforces per-user request rate limits	P1	Completed

ID	Requirement	Priority	Status
FR-RATE-2	Rate limits are tiered: anonymous (10/min), authenticated (100/min), trusted (300/min)	P1	Completed
FR-RATE-3	Action-specific limits apply: 5 posts/hour, 30 comments/hour, 60 votes/min	P1	Completed
FR-RATE-4	Exceeding the limit returns HTTP 429 with a Retry-After header	P1	Completed

1.1.12 1.12 Spam and Abuse Detection (FR-SPAM)

ID	Requirement	Priority	Status
FR-SPAM-1	Content is checked against a keyword blocklist before publishing	P2	Completed
FR-SPAM-2	URLs in posts are checked against a known-bad domain list	P2	Completed
FR-SPAM-3	Duplicate content from the same user is rejected	P2	Completed
FR-SPAM-4	New accounts (< 24h old) cannot create posts; accounts < 1h old cannot comment	P1	Completed
FR-SPAM-5	Asynchronous analysis detects vote manipulation patterns	P3	Completed
FR-SPAM-6	Shadow-banning is available as a moderation action	P3	Completed

1.2 2. Non-Functional Requirements

1.2.1 2.1 Performance (NFR-PERF)

ID	Requirement	Status
NFR-PERF-1	API response time for read operations (feed, post, comments) should be under 200ms at the 95th percentile	Completed
NFR-PERF-2	Vote processing latency from click to updated score display should be under 500ms	Completed
NFR-PERF-3	The system should handle at least 10,000 concurrent users without degradation	Completed
NFR-PERF-4	WebSocket notification delivery should occur within 1 second of the triggering event	Completed
NFR-PERF-5	Search queries should return results within 300ms	Completed

1.2.2 2.2 Scalability (NFR-SCAL)

ID	Requirement	Status
NFR-SCAL-1	The Post Service database is sharded using consistent hashing on <code>community_id</code>	Completed
NFR-SCAL-2	New shards can be added with minimal data migration (~1/N of total data)	Completed
NFR-SCAL-3	Each microservice can be independently scaled via Kubernetes HPA	Completed
NFR-SCAL-4	Kafka consumer groups allow parallel event processing per service	Completed
NFR-SCAL-5	Redis caching reduces direct database load for hot data (community metadata, feeds, vote counts)	Completed

1.2.3 2.3 Security (NFR-SEC)

ID	Requirement	Status
NFR-SEC-1	All external traffic is encrypted with TLS; internal traffic uses mTLS via Istio	Completed

ID	Requirement	Status
NFR-SEC-2	Passwords are hashed with argon2id	Completed
NFR-SEC-3	Emails are encrypted at rest with AES-256	Completed
NFR-SEC-4	IP addresses are never stored in application databases; hashed IPs for abuse detection expire after 24 hours	Completed
NFR-SEC-5	JWT access tokens expire after 15 minutes; refresh tokens after 7 days	Completed
NFR-SEC-6	CORS is restricted to the frontend origin; CSRF tokens protect state-changing requests	Completed
NFR-SEC-7	All database queries use parameterized statements (no string concatenation)	Completed
NFR-SEC-8	Secrets are managed through Kubernetes Secrets or HashiCorp Vault	Completed

1.2.4 2.4 Anonymity and Privacy (NFR-PRIV)

ID	Requirement	Status
NFR-PRIV-1	Other users can only see a user's username, karma, and cake day	Completed
NFR-PRIV-2	Email, authentication method, and IP are never exposed through any API endpoint	Completed
NFR-PRIV-3	Account deletion removes all PII; posts and comments are replaced with [deleted]; vote records are anonymized	Completed
NFR-PRIV-4	Anonymous posting hides the username from all users except community moderators	Completed
NFR-PRIV-5	Raw IP addresses in request logs are rotated and deleted after 7 days	Completed

1.2.5 2.5 Reliability and Availability (NFR-REL)

ID	Requirement	Status
NFR-REL-1	The system targets 99.9% uptime (roughly 8.7 hours of downtime per year)	Completed
NFR-REL-2	Each service has readiness and liveness probes; Kubernetes restarts unhealthy pods automatically	Completed
NFR-REL-3	Circuit breakers (sony/gobreaker) prevent cascade failures between services	Completed
NFR-REL-4	Kafka provides durability for events; if a consumer goes down, it resumes from the last committed offset	Completed
NFR-REL-5	PostgreSQL instances have automated daily backups with point-in-time recovery	Completed

1.2.6 2.6 Usability (NFR-USE)

ID	Requirement	Status
NFR-USE-1	Page load time should be under 2 seconds on a fast 4G connection	Completed
NFR-USE-2	The Astro frontend ships minimal JavaScript; interactive elements load as Svelte islands	Completed
NFR-USE-3	The application is responsive and usable on mobile, tablet, and desktop	Completed
NFR-USE-4	Error messages are specific (e.g., "Username already taken" rather than "Registration failed")	Completed

1.2.7 2.7 Observability (NFR-OBS)

ID	Requirement	Status
NFR-OBS-1	Prometheus collects metrics from every service; Grafana provides per-service dashboards	Completed
NFR-OBS-2	All services emit structured JSON logs, aggregated in Loki and queryable through Grafana	Completed

ID	Requirement	Status
NFR-OBS-3	OpenTelemetry traces span across service boundaries; Jaeger visualizes end-to-end request flows	Completed
NFR-OBS-4	Grafana alerts fire on error rate spikes, P99 latency exceeding thresholds, pod restarts, and Kafka consumer lag	Completed

1.3 3. Acceptance Criteria

ID	Criterion	Status
AC-1	A user can register with email/password, verify via OTP, and log in	Completed
AC-2	A user can register via Google OAuth and choose a username	Completed
AC-3	A user can create a community and another user can join it	Completed
AC-4	A user can create a text post in a community and it appears in the feed	Completed
AC-5	A user can comment on a post and reply to existing comments (at least 3 levels deep)	Completed
AC-6	Upvoting/downvoting updates the score and the author's karma	Completed
AC-7	The home feed aggregates posts from joined communities, sorted by Hot/New/Top	Completed
AC-8	Search returns relevant posts by title and body within 300ms	Completed
AC-9	A WebSocket notification is received within 1 second when someone replies to your post	Completed
AC-10	Rate limiting returns HTTP 429 when the threshold is exceeded	Completed

ID	Criterion	Status
AC-11	A moderator can remove a post, ban a user, and pin a post in their community	Completed
AC-12	Account deletion wipes PII and replaces content with [deleted]	Completed
AC-13	The post database is sharded; adding a shard migrates only a fraction of the data	Completed
AC-14	All services run on Kubernetes with health checks and auto-scaling	Completed

1.4 4. Summary

Category	Total	Completed	Pending
Functional Requirements (FR)	75	75	0
Non-Functional Requirements (NFR)	36	36	0
Acceptance Criteria (AC)	14	14	0
Grand Total	125	125	0

All promised features and requirements have been successfully implemented.